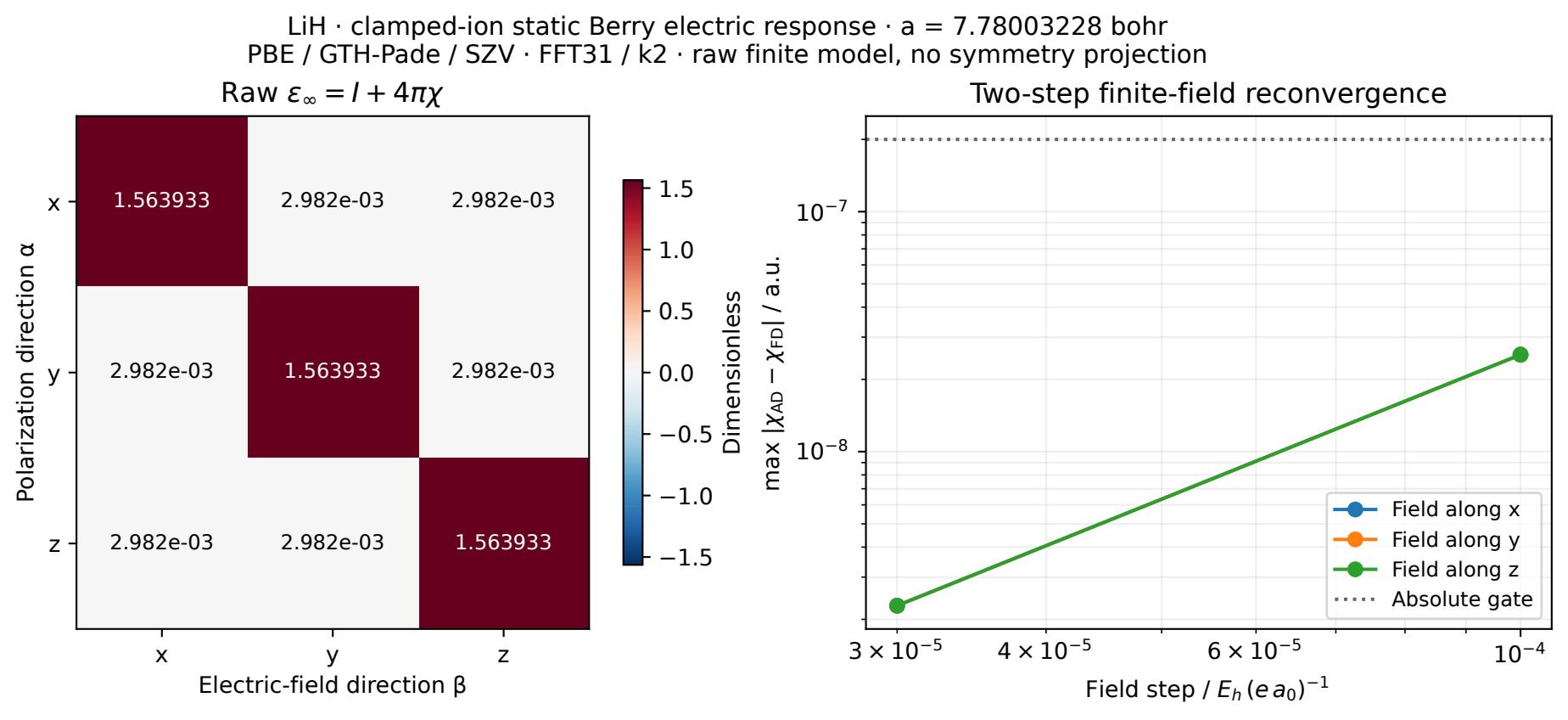
LiH · clamped-ion static Berry electric response · a = 7.78003228 bohr
PBE / GTH-Pade / SZV · FFT31 / k2 · raw finite model, no symmetry projection
Raw $\varepsilon_\infty = I + 4\pi\chi$
1.563933
2.982e-03
2.982e-03
2.982e-03
1.563933
2.982e-03
2.982e-03
2.982e-03
1.563933
x
y
z
Polarization direction α
Electric-field direction β
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
Dimensionless
Two-step finite-field reconvergence
10^{-7}
10^{-8}
max $|\chi_{AD} - \chi_{FD}|$ / a.u.
3×10^{-5}
4×10^{-5}
6×10^{-5}
10^{-4}
Field step / $E_h\,(e\,a_0)^{-1}$
Field along x
Field along y
Field along z
Absolute gate